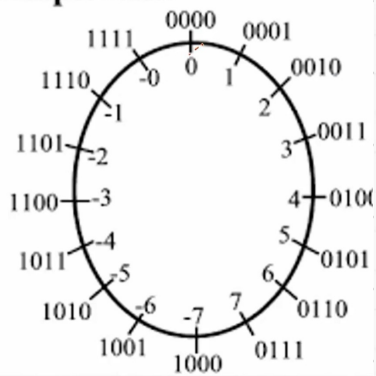
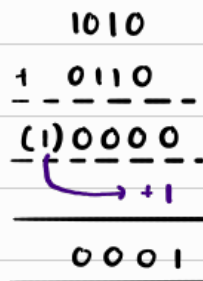


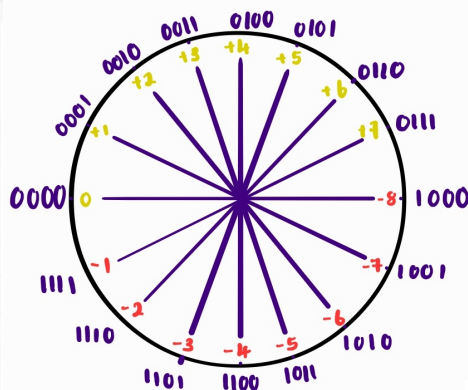
Ones complement



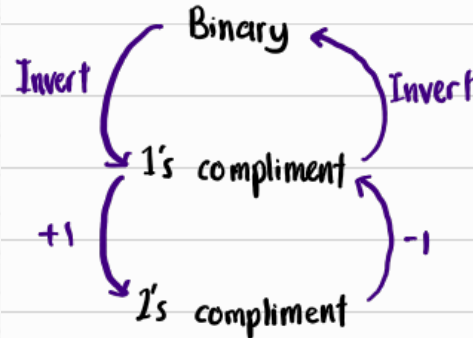
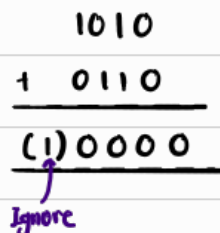
$$-(2^{n-1}) - 1 \leq K \leq 2^{n-1} - 1$$



2's Complement System



$$-2^{n-1} \leq K \leq 2^{n-1} - 1$$



2	25
2	12
2	6
2	3
2	1
	0

- 1 ← First remainder
- 0 ← Second Remainder
- 0 ← Third Remainder
- 1 ← Fourth Remainder
- 1 ← Fifth Remainder

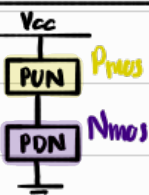
Read Up

Binary Number = 11001

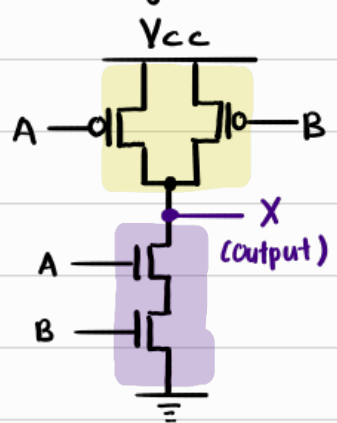
Circuit Globe

CMOS Logic Gate

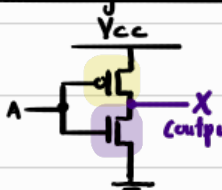
Pmos ON at 0
Nmos ON at 1



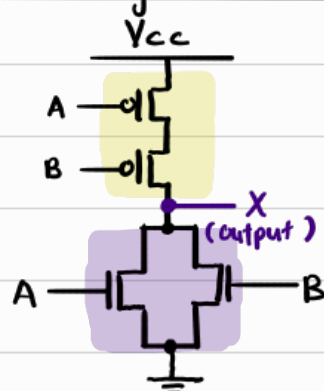
NAND gate [A · B]



NOT gate [A]



NOR gate [A + B]



PUN

NAND ~ Parallel

NOR ~ Series

PDN

NAND ~ Series

NOR ~ Parallel

OR (+)

Theorem

And (-)

$$X + 0 = X$$

Identity

$$X \cdot 1 = X$$

$$X + 1 = 1$$

Null

$$X \cdot 0 = 0$$

$$X + X = X$$

Idempotent

$$X \cdot X = X$$

$$\bar{\bar{X}} = X$$

Involution

$$\bar{\bar{X}} = X$$

$$X + \bar{X} = 1$$

Complementarity

$$X \cdot \bar{X} = 0$$

$$(X \cdot Y) + (X \cdot \bar{Y}) = X$$

Uniting

$$(X + Y) \cdot (X + \bar{Y}) = X$$

$$X + XY = X$$

Absorption

$$(X \cdot \bar{Y}) + Y = X + Y$$

$$XY + YZ + \bar{X}Z = XY + \bar{X}Z$$

Consensus

$$(X + Y) \cdot (Y + Z) \cdot (X + Z) = (X + Y) \cdot (X + Z)$$

$$\overline{X + Y} = \bar{X} \cdot \bar{Y}$$

DeMorgan

$$\overline{XY} = \bar{X} + \bar{Y}$$

Buffer



Input	Output
0	0
1	1

AND



A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

OR



A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

XOR



A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

Inverter



Input	Output
0	1
1	0

NAND



A	B	Output
0	0	1
1	0	1
0	1	1
1	1	0

NOR



A	B	Output
0	0	1
1	0	0
0	1	0
1	1	0

XNOR



A	B	Output
0	0	1
1	0	0
0	1	0
1	1	1

$$AB + \bar{A}\bar{B}$$

HEXADECIMAL VALUES

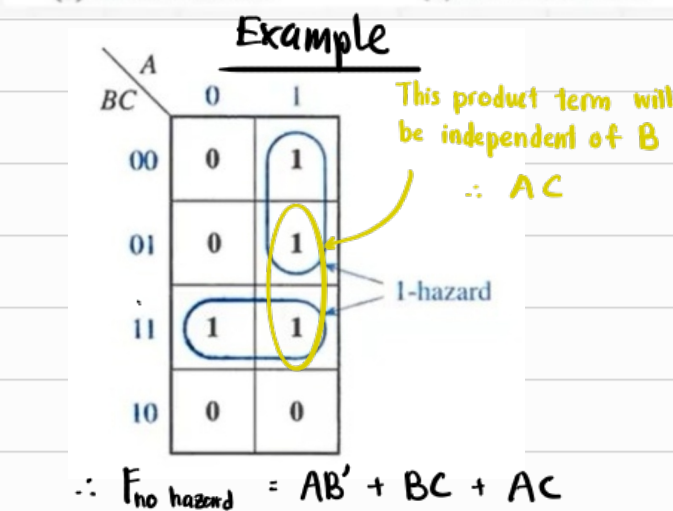
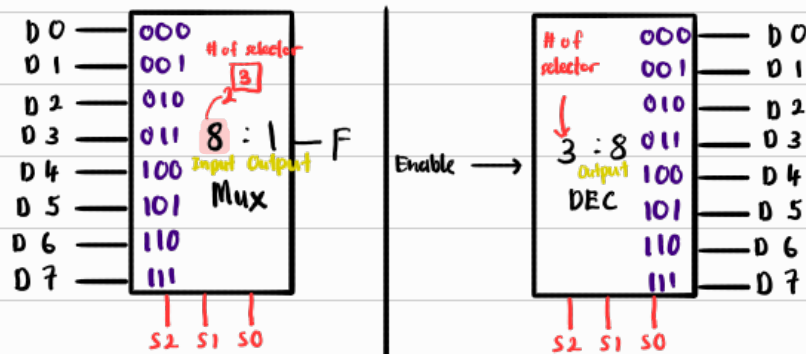
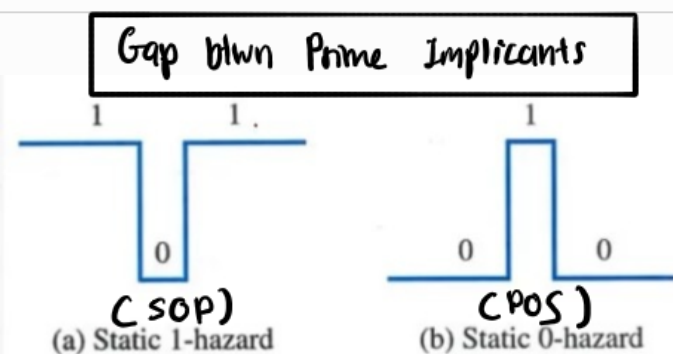
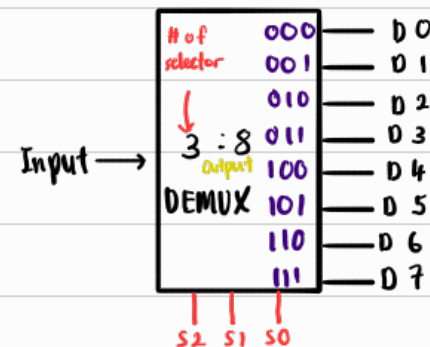
DECIMAL RANGE

00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	00-15
10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F	16-31
20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F	32-47
30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F	48-63
40	41	42	43	44	45	46	47	48	49	4A	4B	4C	4D	4E	4F	64-79
50	51	52	53	54	55	56	57	58	59	5A	5B	5C	5D	5E	5F	80-95
60	61	62	63	64	65	66	67	68	69	6A	6B	6C	6D	6E	6F	96-111
70	71	72	73	74	75	76	77	78	79	7A	7B	7C	7D	7E	7F	112-127
80	81	82	83	84	85	86	87	88	89	8A	8B	8C	8D	8E	8F	128-143
90	91	92	93	94	95	96	97	98	99	9A	9B	9C	9D	9E	9F	144-159
A0	A1	A2	A3	A4	A5	A6	A7	A8	A9	AA	AB	AC	AD	AE	AF	160-175
B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	BA	BB	BC	BD	BE	BF	176-191
C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	CA	CB	CC	CD	CE	CF	192-207
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	DA	DB	DC	DD	DE	DF	208-223
E0	E1	E2	E3	E4	E5	E6	E7	E8	E9	EA	EB	EC	ED	EE	EF	224-239
F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	FA	FB	FC	FD	FE	FF	240-255

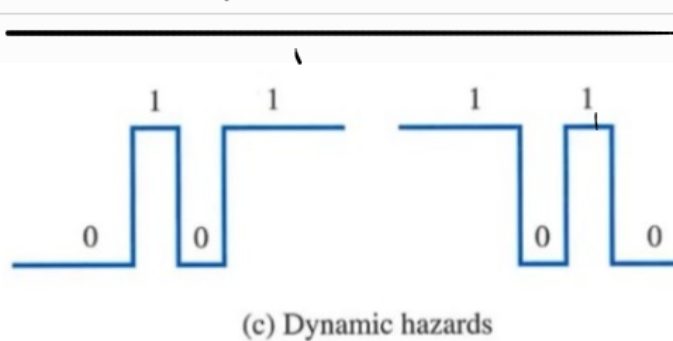
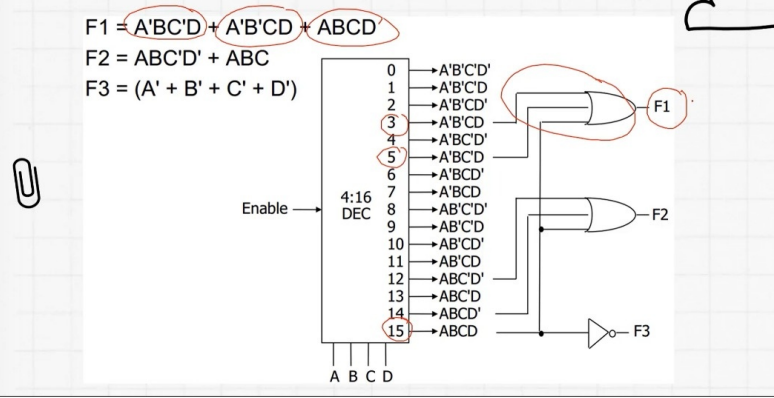
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Decimal Number	Binary Number	Decimal Number	Binary Number
1	1	11	1011
2	10	12	1100
3	11	13	1101
4	100	14	1110
5	101	15	1111
6	110	16	10000
7	111	17	10001
8	1000	18	10010
9	1001	19	10011
10	1010	20	10100

Karnaugh Map Terminology			
Term	Definition	Example (3 variables: A, B, C)	In K-map
Minterm	A product (AND) term where all variables appear once (complemented or not). Represents a single 1-cell in the map.	$A'B'C \rightarrow$ true only when $A=0, B=0, C=1$	One cell marked 1
Maxterm	A sum (OR) term where all variables appear once (complemented or not). Represents a single 0-cell in the map.	$(A + B + C') \rightarrow$ false only when $A=0, B=0, C=1$	One cell marked 0
Implicant	A product term that covers one or more minterms. In the map, any group of 1s forming a rectangle of size 2^n .	Grouping $AB'C$ and $AB'C' \rightarrow$ implicant = AB'	A block of 1s
Prime Implicant	An implicant that cannot be combined into a larger group. (The largest possible block of 1s.)	A 4-cell group $\rightarrow A$	Biggest rectangle of 1s
Essential Prime Implicant	A prime implicant that covers at least one unique minterm not covered by any other. Must be included in final simplification.	If only one block covers $A'B'C$, that block is essential	Highlighted must-have block
Redundant Implicant	An implicant that doesn't add coverage (all its 1s already covered by others). Can be omitted.	A 2-cell group fully inside a 4-cell group	Extra, unnecessary grouping
Don't-care (X)	Input combos that are irrelevant or impossible. Can be treated as 0 or 1 to aid simplification.	$XYZ=111$ never happens $\rightarrow$ mark cell as 'X'	Flexible cell



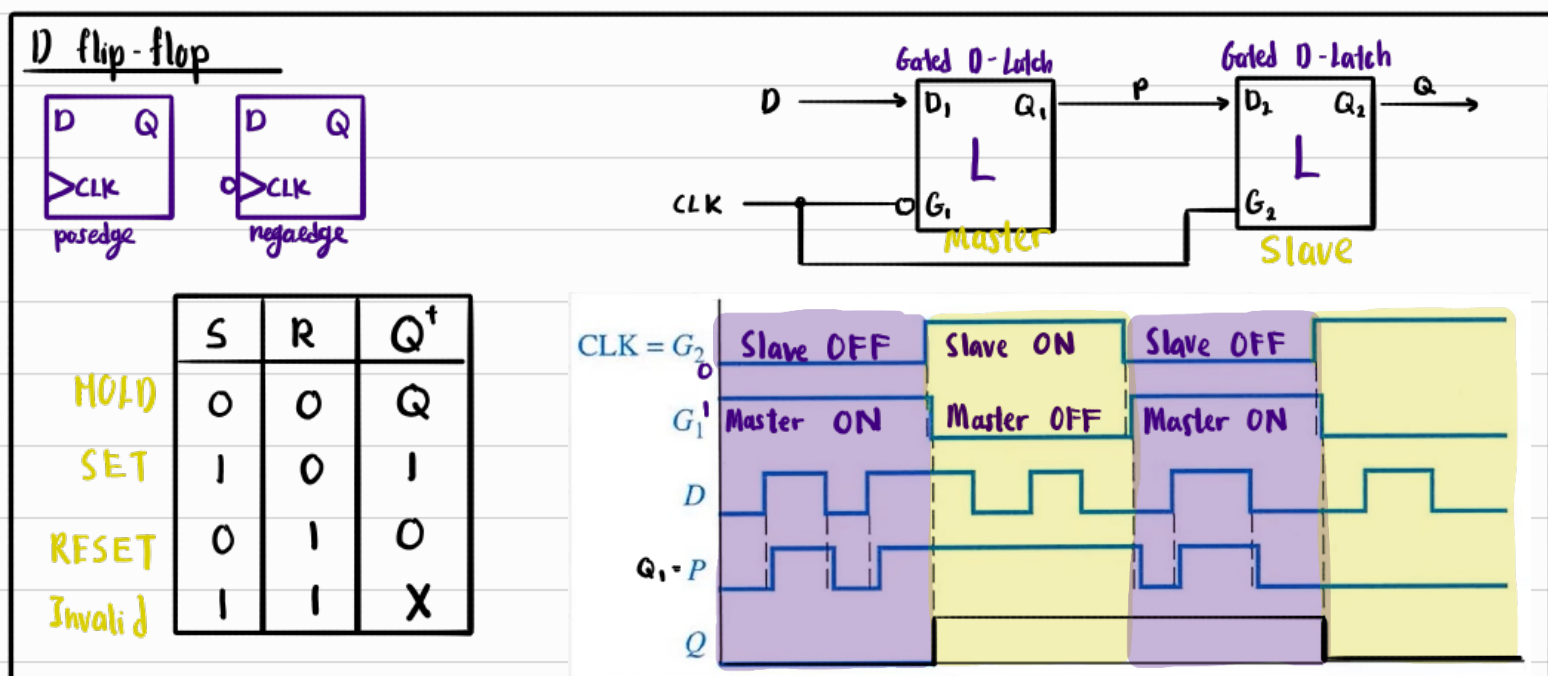
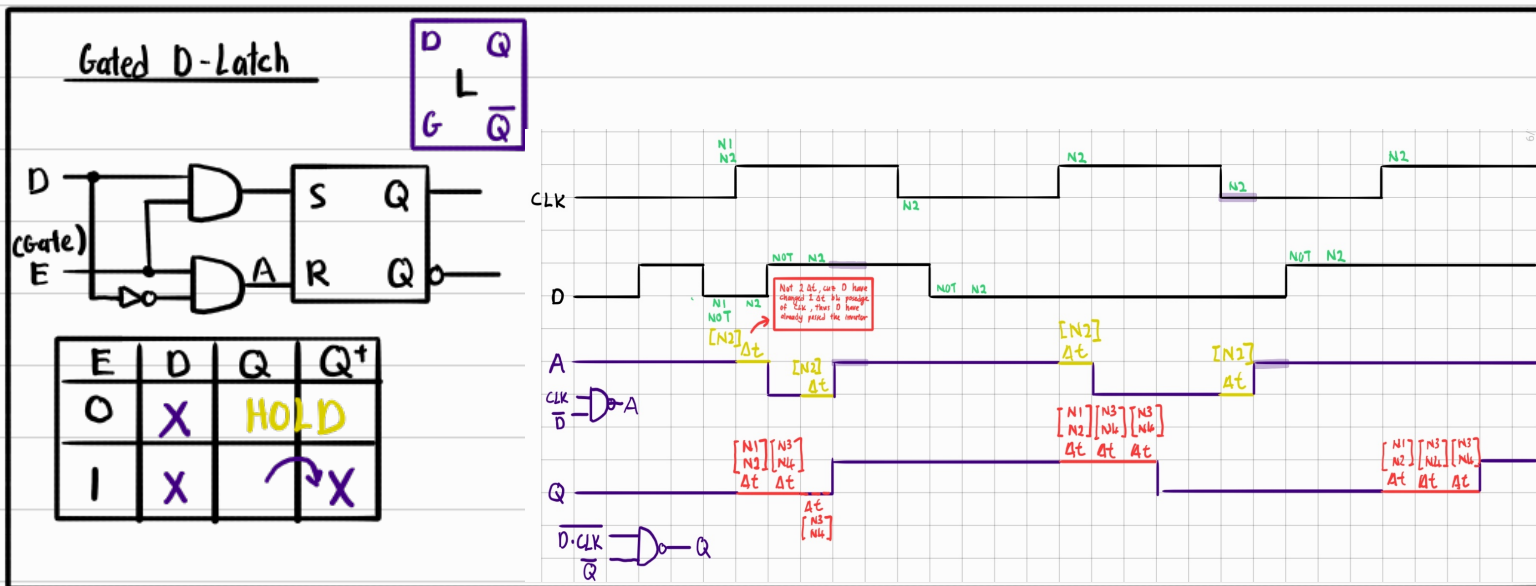
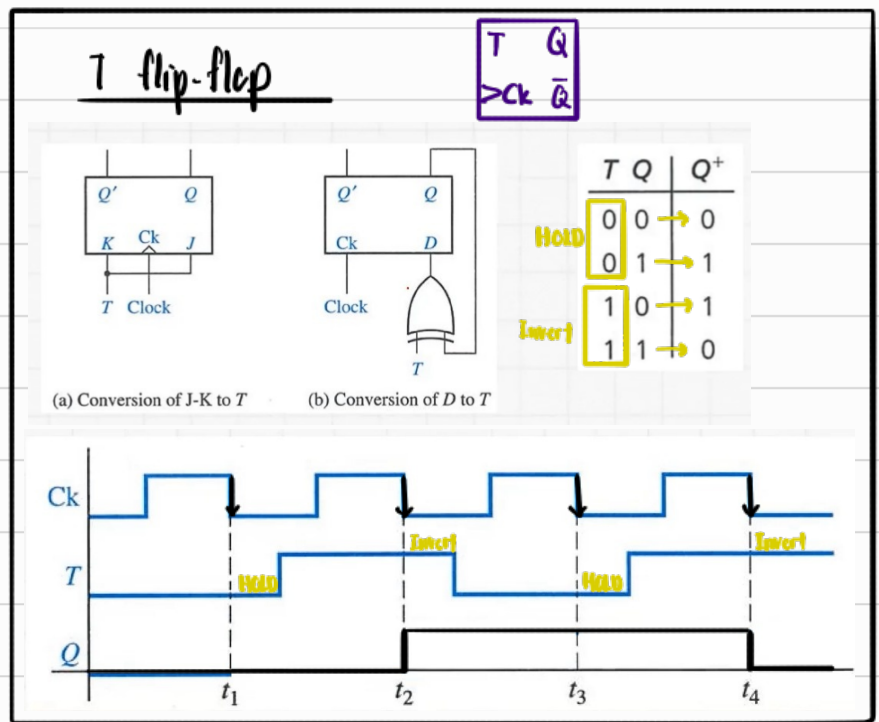
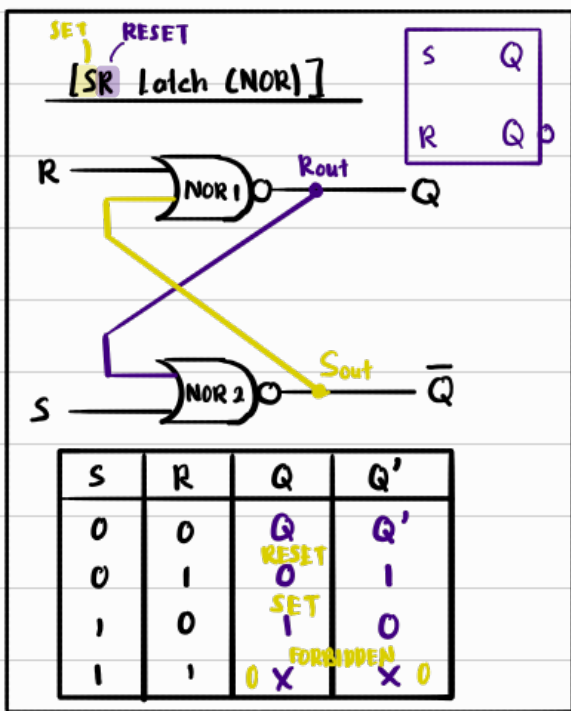
Decoders as general logic



Question W4Q4

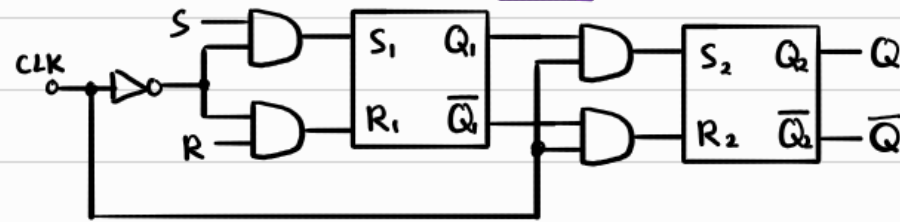
Implement the following using a 4-to-1 multiplexer where B must be one of the selector bits.

A	B	C	D	Q
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0



SR Flip-flop

S	Q
$\triangleright \text{CK}$	
R	Q'



At pos-edge

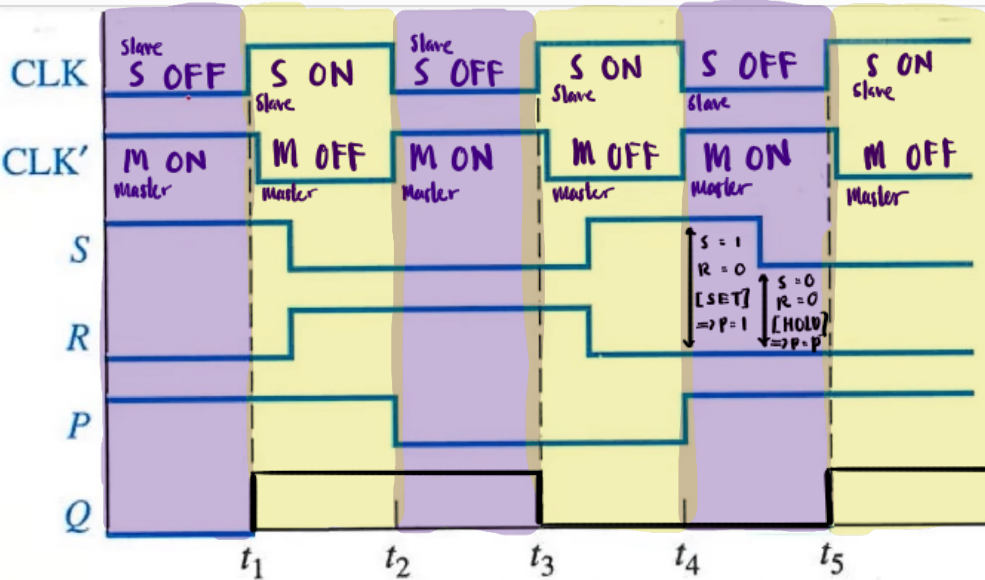
S	R	Q^+
0	0	Q
1	0	1
0	1	0
1	1	X

HOLD

SET

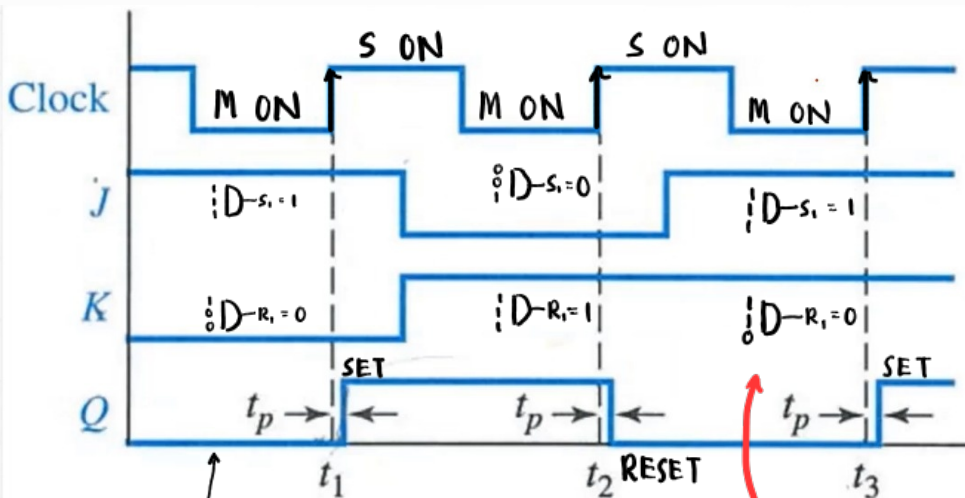
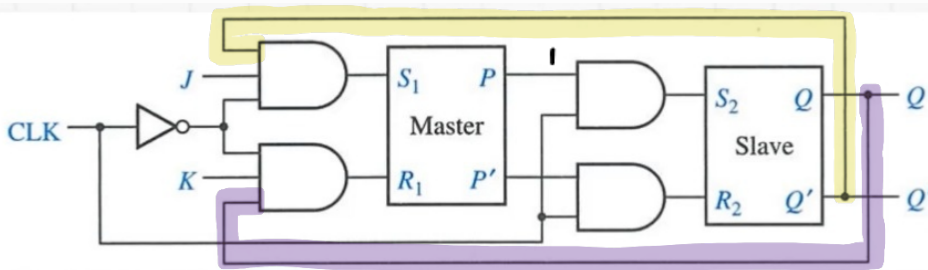
RESET

Invalid



JK flip-flop

J	Q
$\triangleright \text{CK}$	
K	Q'



J	K	Q^+	Q'
0	0	Q	Q'
0	1	0	1
1	0	1	0
1	1	Q'	Q

Assuming Q initially at 0

Notice that $J=1, K=1$ is not forbidden, because $Q=0$ is used as an input.

Example 1

Design a sequential logic design that generates the following output sequence using JK flip-flops

00, 01, 11, 10, 00

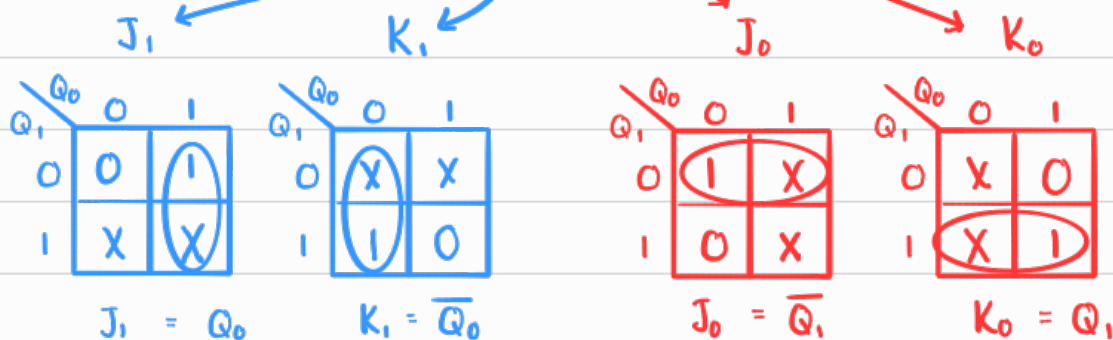
	J	K	Q^+
HOLD	0	0	Q
SET	1	0	1
RESET	0	1	0
Toggle	1	1	Q'

Since we need a 2-bit output, we need 2 JK flip-flops

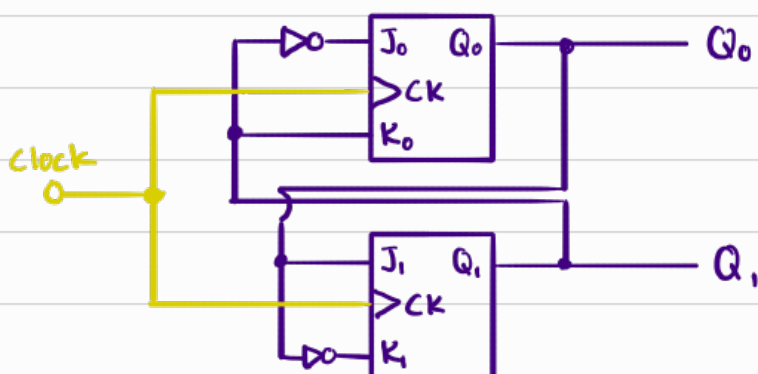
① State transition table

Current Q_1, Q_0	Next Q_1^+, Q_0^+	J, K	J_0, K_0
0 0	0 1	HOLD/RESET 0 X	SET / Toggle 1 X
0 1	1 1	SET / Toggle 1 X	HOLD / SET X 0
1 1	1 0	HOLD / SET X 0	RESET / Toggle X 1
1 0	0 0	RESET / Toggle X 1	HOLD / RESET 0 X

② K-maps



③ Implementation



Example 1

Design a 2-bit counter using 1 flip-flops. If input X is 0, counter increase by 1, while if the input X is 1, counter decrease by 1.

At pos-edge

T	Q^+
0	Q
1	Q'

HOLD
Toggle

Since we need a 2-bit output, we need 2 JK flip-flops

① State transition table

3 variable
Input $\therefore 2^3$ combination

X	$Q_1 Q_0$	$Q_1^+ Q_0^+$	T_1	T_0
0	0 0	0 1	0	1
1	0 0	1 1	1	1
0	0 1	1 0	1	1
1	0 1	0 0	0	1
0	1 0	1 1	0	1
1	1 0	0 1	1	1
0	1 1	0 0	1	1
1	1 1	1 0	0	1

② K-map

T_1

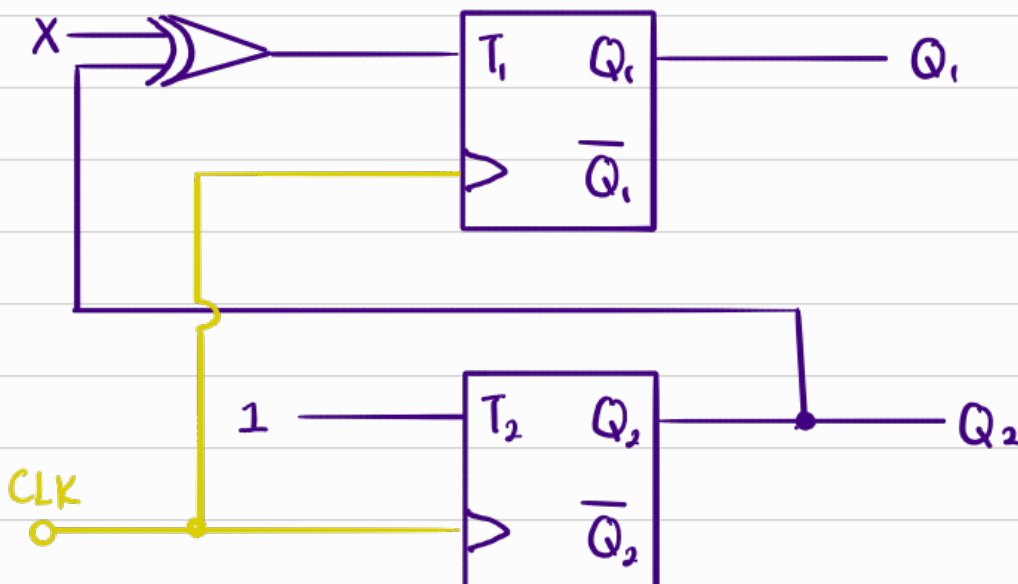
	$Q_1 Q_0$	00	01	11	10
X	0	0	1	1	0
1	1	1	0	0	1

$$T_1 = \bar{X}Q_0 + X\bar{Q}_0$$

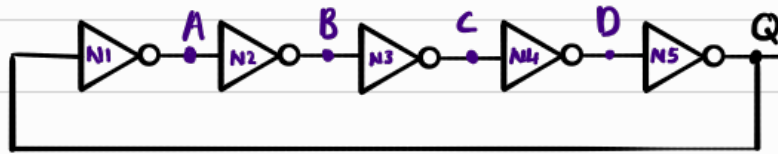
$$T_1 = X \oplus Q_0$$

$$T_0 = 1$$

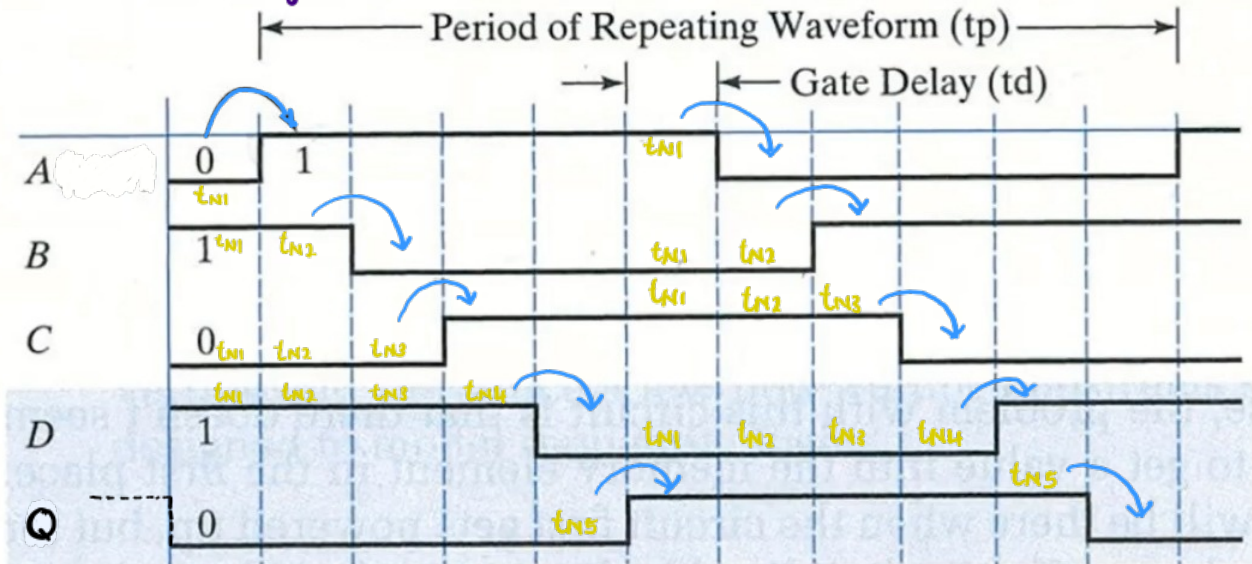
③ Implementation



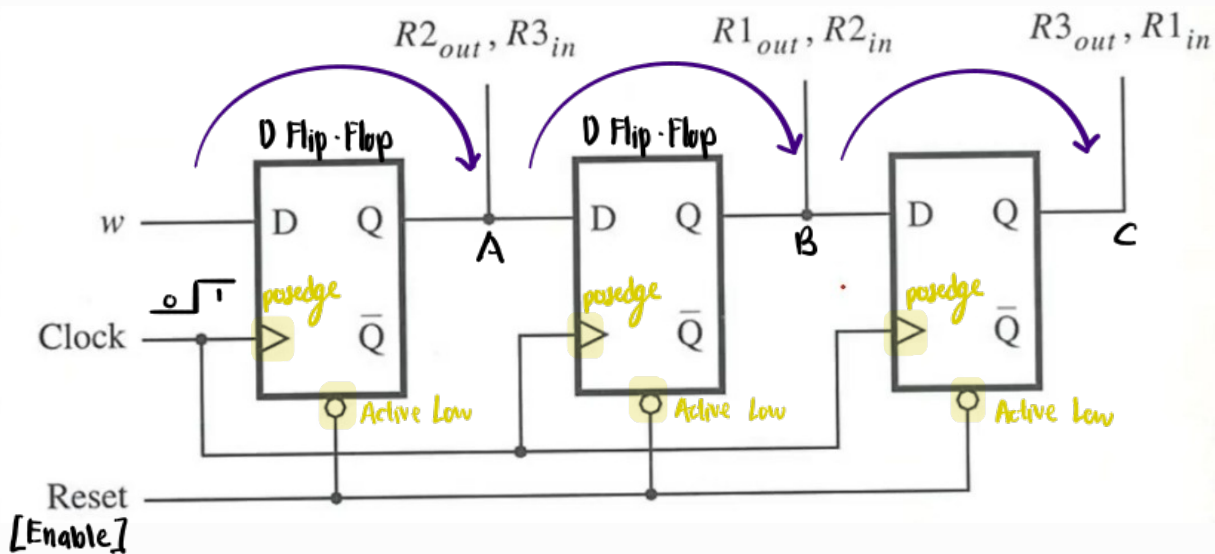
Ring Oscillator



Assume Q changes from 1 to 0 [$A=0, B=1, C=0, D=1$]

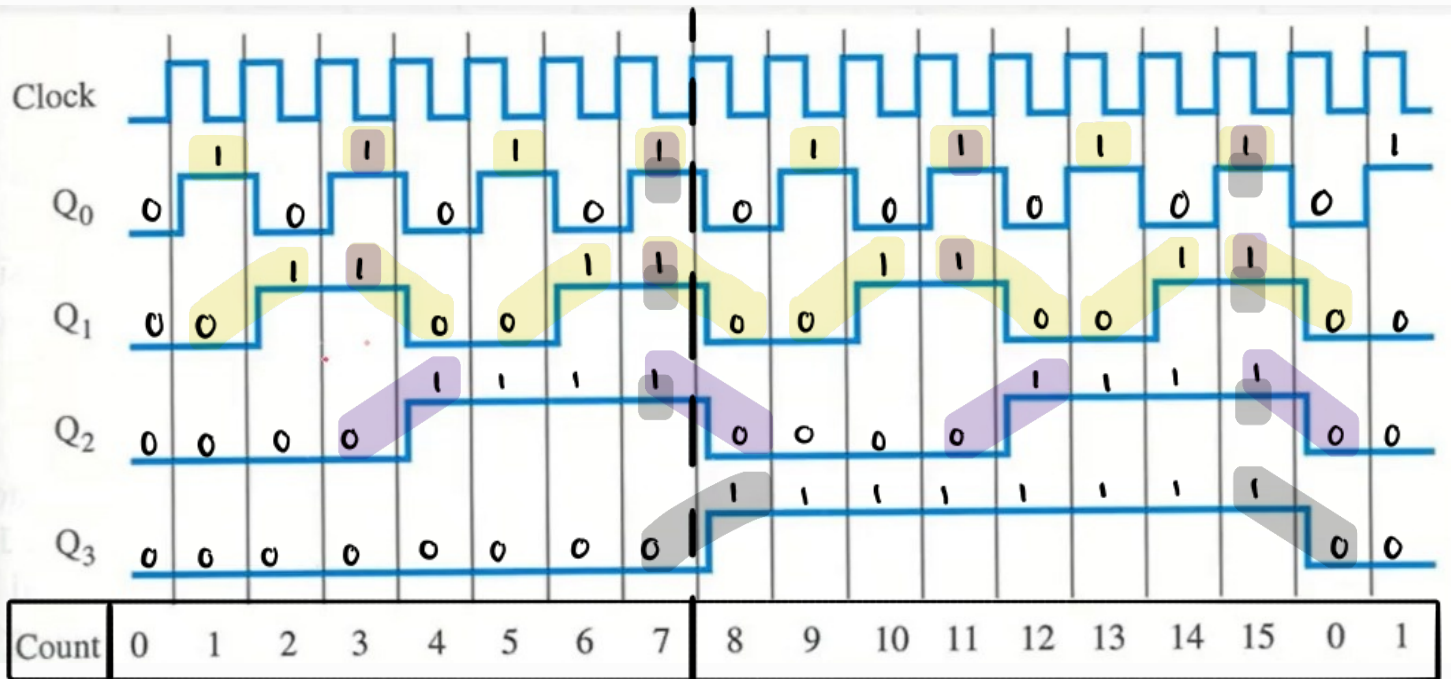
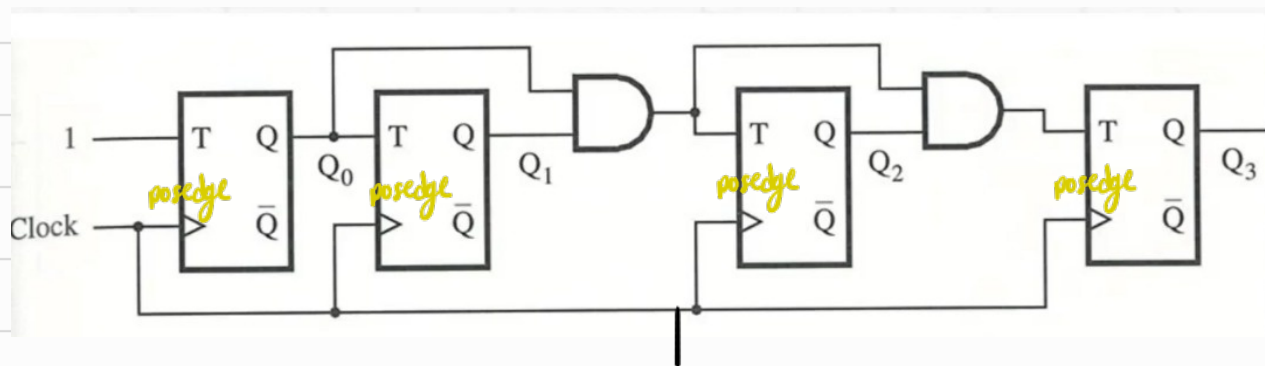


3 bit shift register



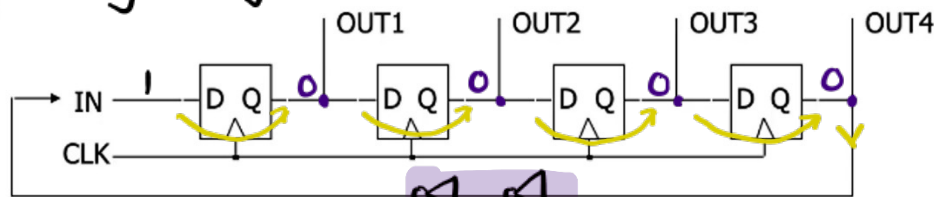
Every positive edge, each D flip-flop transmit $D \rightarrow Q$.
Causing all data in A, B, C to move to the right
by 1 step

4 bit counter using T flip-flops



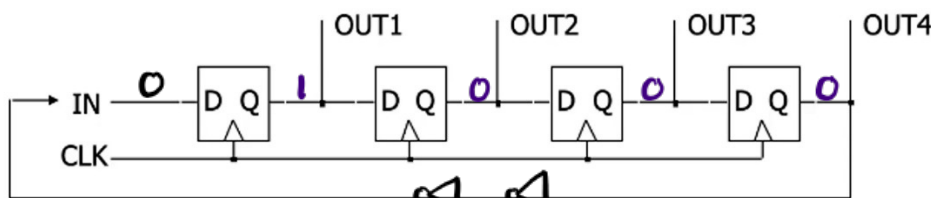
One-hot sequence generator

Assuming initially $IN = 1$

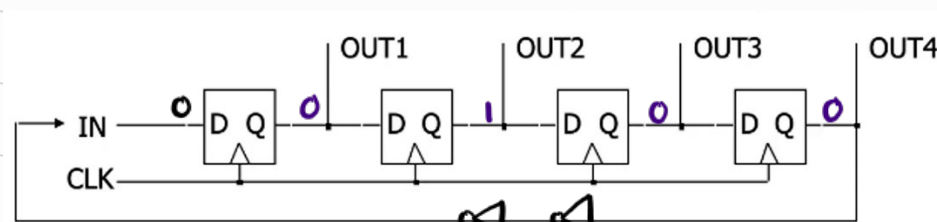


0 0 0 0

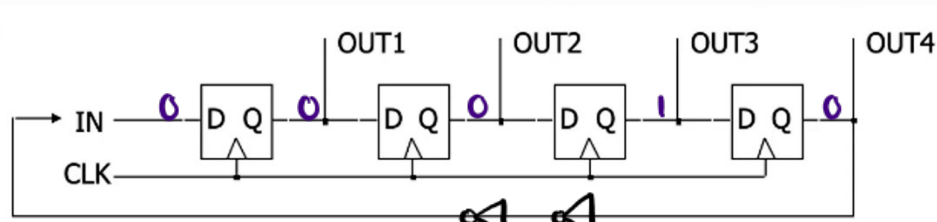
To avoid OUT4 & IN from short-circuiting [Introduce Delay]



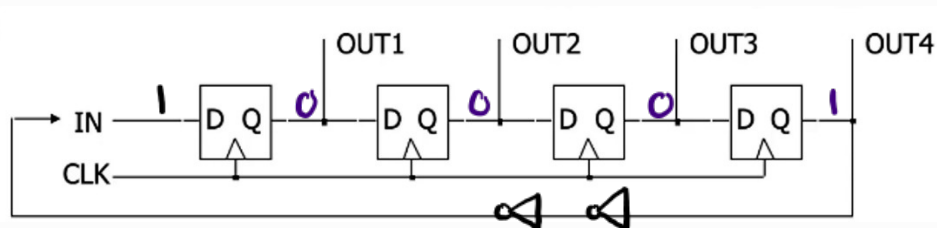
1 0 0 0



0 1 0 0



0 0 1 0



0 0 0 1